

1704. Determine if String Halves Are Alike

Solved

Easy Topics Companies Hint

You are given a string `s` of even length. Split this string into two halves of equal lengths, and let `a` be the first half and `b` be the second half.

Two strings are **alike** if they have the same number of vowels (`'a'`, `'e'`, `'i'`, `'o'`, `'u'`, `'A'`, `'E'`, `'I'`, `'O'`, `'U'`). Notice that `s` contains uppercase and lowercase letters.

Return `true` if `a` and `b` are alike. Otherwise, return `false`.

Example 1:

Input: `s = "book"`
Output: `true`
Explanation: `a = "bo"` and `b = "ok"`. `a` has 1 vowel and `b` has 1 vowel. Therefore, they are alike.

Example 2:

Input: `s = "textbook"`
Output: `false`
Explanation: `a = "text"` and `b = "book"`. `a` has 1 vowel whereas `b` has 2. Therefore, they are not alike. Notice that the vowel `o` is counted twice.

Constraints:

- `2 <= s.length <= 1000`
- `s.length` is even.
- `s` consists of **uppercase and lowercase** letters.

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Java Auto

```
1 class Solution {
2     public boolean halvesAreAlike(String s) {
3         int n = s.length();
4         int count = 0;
5
6         for (int i = 0; i < n / 2; i++) {
7             if (isVowel(s.charAt(i))) {
8                 count++;
9             }
10        }
11
12        for (int i = n / 2; i < n; i++) {
13            if (isVowel(s.charAt(i))) {
14                count--;
15            }
16        }
17
18        return count == 0;
```

Saved

Ln 27, Col 2

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input

`s =`
`"book"`

Output

`true`

Expected

`true`

Lapindromes

Lapindrome is defined as a string which when split in the middle, gives two halves having the same characters and same frequency of each character. If there are odd number of characters in the string, we ignore the middle character and check for lapindrome. For example **gaga** is a lapindrome, since the two halves **ga** and **ga** have the same characters with same frequency. Also, **abccab**, **rotor** and **xyzxy** are a few examples of lapindromes. Note that **abbaab** is NOT a lapindrome. The two halves contain the same characters but their frequencies do not match.

Your task is simple. Given a string, you need to tell if it is a lapindrome.

Input:

First line of input contains a single integer **T**, the number of test cases.

Each test is a single line containing a string **S** composed of only lowercase English alphabet.

Output:

For each test case, output on a separate line: "YES" if the string is a lapindrome and "NO" if it is not.

Constraints:

$$1 \leq T \leq 100$$

$$2 \leq |S| \leq 1000, \text{ where } |S| \text{ denotes the length of } S$$

Sample 1:

Input	Output
6	YES
gaga	NO
abcde	YES

rotor

Status : Successfully executed

Time:

0.0700 secs

Memory:

42.024 Mb

Sample Input

```
6
gaga
abcde
rotor
xyzxy
abbaab
ababc
```

Your Output

```
YES
NO
YES
YES
NO
NO
```

Visualize Code

Run

Alice and Bob each created one problem for HackerRank. A reviewer rates the two challenges, awarding points on a scale from 1 to 100 for three categories: problem clarity, originality, and difficulty.

The rating for Alice's challenge is the triplet $a = (a[0], a[1], a[2])$, and the rating for Bob's challenge is the triplet $b = (b[0], b[1], b[2])$.

The task is to calculate their comparison points by comparing each category:

- If $a[i] > b[i]$, then Alice is awarded 1 point.
- If $a[i] < b[i]$, then Bob is awarded 1 point.
- If $a[i] = b[i]$, then neither person receives a point.

Example

$a = [1, 2, 3]$

$b = [3, 2, 1]$

- For elements $a[0]$, Bob is awarded a point because $a[0] < b[0]$.
- For the equal elements $a[1]$ and $b[1]$, no points are earned.
- Finally, for elements 2, $a[2] > b[2]$ so Alice receives a point.

The return array is $[1, 1]$ with Alice's score first and Bob's second.

Function Description

Complete the function `compareTriplets` with the following parameter(s):

- `int a[3]`: Alice's challenge rating
- `int b[3]`: Bob's challenge rating

Returns

- `int[2]`: the first element is Alice's score and the second is Bob's score

```

15
16 // Read Bob's ratings
17 for (int i = 0; i < 3; i++) {
18     b[i] = sc.nextInt();
19 }
20
21 int alice = 0;
22 int bob = 0;
23
24 // Compare ratings
25 for (int i = 0; i < 3; i++) {
26     if (a[i] > b[i]) {
27         alice++;
28     } else if (a[i] < b[i]) {
29         bob++;
30     }
31 }
32
33 System.out.println(alice + " " + bob);
34
35 sc.close();
36 }
37 }

```

Line: 11 Col: 32



Upload Code as File

☐

Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.



Sample Test case 0

Input (stdin)

Download

Line: 21 Col: 23

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Congratulations!

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✓ Sample Test case 0

Input (stdin)

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```
1 5 6 7
2 3 6 10
```

Your Output (stdout)

```
1 1 1
```

Expected Output

[Download](#)

```
1 1 1
```

Description Editorial Solutions Submissions

217. Contains Duplicate

Easy Topics Companies

Given an integer array `nums`, return `true` if any value appears **at least twice** in the array, and return `false` if every element is distinct.

Example 1:

Input: `nums = [1,2,3,1]`

Output: `true`

Explanation:

The element 1 occurs at the indices 0 and 3.

Example 2:

Input: `nums = [1,2,3,4]`

Output: `false`

Explanation:

All elements are distinct.

Example 3:

Input: `nums = [1,1,1,3,3,4,3,2,4,2]`

Output: `true`

Constraints:

- $1 \leq \text{nums.length} \leq 10^5$
- $-10^9 \leq \text{nums}[i] \leq 10^9$

14.2K 470

271 Online

Code Note

Java Auto

```
1 import java.util.HashSet;
2
3 class Solution {
4     public boolean containsDuplicate(int[] nums) {
5         HashSet<Integer> set = new HashSet<>();
6
7         for (int num : nums) {
8             if (set.contains(num)) {
9                 return true;
10            }
11            set.add(num);
12        }
13
14        return false;
15    }
16 }
```

Saved

Ln 16, Col 2

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

nums =
[1,2,3,1]

Output

Given a time in **12-hour AM/PM format**, convert it to military (24-hour) time.

Note: - 12:00:00AM on a 12-hour clock is 00:00:00 on a 24-hour clock.

- 12:00:00PM on a 12-hour clock is 12:00:00 on a 24-hour clock.

Example

- $s = '12:01:00PM'$

Return '12:01:00'.

- $s = '12:01:00AM'$

Return '00:01:00'.

Function Description

Complete the *timeConversion* function with the following parameter(s):

- *string s*: a time in **12** hour format

Returns

- *string*: the time in **24** hour format

Input Format

A single string s that represents a time in **12-hour** clock format (i.e.: **hh:mm:ssAM** or **hh:mm:ssPM**).

Constraints

- All input times are valid

Sample Input 0

07:05:45PM

Sample Output 0

19:05:45


```
1  import java.util.*;
2
3  public class Solution {
4
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7
8          String s = sc.next();
9
10         int hour = Integer.parseInt(s.substring(0, 2));
11         String period = s.substring(8, 10);
12
13         if (period.equals("AM")) {
14             if (hour == 12) {
15                 hour = 0;
16             }
17         } else {
18             if (hour != 12) {
19                 hour += 12;
20             }
21         }
22
23         System.out.printf("%02d%s\n", hour, s.substring(2, 8));
24
25         sc.close();
26     }
27 }
```


 Upload Code as File

☐ Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

1 07:05:45PM

Download

Your Output (stdout)

1 19:05:45

Expected Output

1 19:05:45

Download

283. Move Zeroes

Easy Topics Companies Hint

Given an integer array `nums`, move all `0`'s to the end of it while maintaining the relative order of the non-zero elements.

Note that you must do this in-place without making a copy of the array.

Example 1:

Input: `nums = [0,1,0,3,12]`
Output: `[1,3,12,0,0]`

Example 2:

Input: `nums = [0]`
Output: `[0]`

Constraints:

- $1 \leq \text{nums.length} \leq 10^4$
- $-2^{31} \leq \text{nums}[i] \leq 2^{31} - 1$

Follow up: Could you minimize the total number of operations done?

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Yes No

Accepted 5 578 094 10 784 Acceptance Rate 64.3%

19.8K 451 261 Online

Code Note X

Java Auto

```
1 class Solution {
2     public void moveZeroes(int[] nums)
3     {
4         int index = 0;
5
6         // Move all non-zero elements to the front
7         for (int i = 0; i < nums.length; i++) {
8             if (nums[i] != 0) {
9                 nums[index] = nums[i];
10                index++;
11            }
12        }
13
14        // Fill the remaining positions with zero
15        while (index < nums.length) {
16            nums[index] = 0;
17            index++;
18        }
19    }
20 }
```

Saved Ln 20, Col 8

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input
nums =
[0,1,0,3,12]
Output

Given a square matrix, calculate the absolute difference between the sums of its diagonals.

For example, the square matrix *arr* is shown below:

```
1 2 3
4 5 6
9 8 9
```

- The left-to-right diagonal = $1 + 5 + 9 = 15$.
- The right-to-left diagonal = $3 + 5 + 9 = 17$.

Their absolute difference is $|15 - 17| = 2$.

Function description

Complete the *diagonalDifference* function with the following parameter:

- *int arr[n][m]*: a 2-D array of integers

Return

- *int*: the absolute difference in sums along the diagonals

Input Format

The first line contains a single integer, *n*, the number of rows and columns in the square matrix *arr*.

Each of the next *n* lines describes a row, *arr[i]*, and consists of *n* space-separated integers *arr[i][j]*.

Constraints

- $-100 \leq arr[i][j] \leq 100$

Sample Input

STDIN	Function
-----	-----
3	arr[][] sizes n = 3, m = 3
11 2 4	arr = [[11, 2, 4], [4, 5, 6], [10, 8, -12]]
4 5 6	
10 8 -12	

```
2
3 ✓ public class Solution {
4
5 ✓     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         int n = sc.nextInt();
9
10        int[][] arr = new int[n][n];
11
12        for (int i = 0; i < n; i++) {
13            for (int j = 0; j < n; j++) {
14                arr[i][j] = sc.nextInt();
15            }
16        }
17
18        int diagonal1 = 0;
19        int diagonal2 = 0;
20
21 ✓    for (int i = 0; i < n; i++) {
22        diagonal1 += arr[i][i];
23        diagonal2 += arr[i][n - 1 - i];
24    }
25
26    System.out.println(Math.abs(diagonal1 - diagonal2));
27
28    sc.close();
29 }
30 }
31
```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1 3
2 11 2 4
3 4 5 6
4 10 8 -12
```

[Download](#)

Your Output (stdout)

```
1 15
```

Expected Output

```
1 15
```

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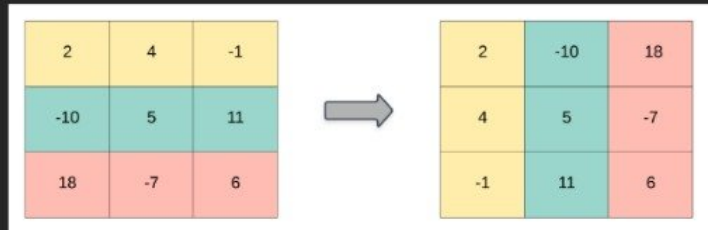
Description Editorial Solutions Submissions

867. Transpose Matrix

Easy Topics Companies Hint

Given a 2D integer array `matrix`, return the *transpose* of `matrix`.

The *transpose* of a matrix is the matrix flipped over its main diagonal, switching the matrix's row and column indices.



Example 1:

Input: `matrix = [[1,2,3],[4,5,6],[7,8,9]]`

Output: `[[1,4,7],[2,5,8],[3,6,9]]`

Example 2:

Input: `matrix = [[1,2,3],[4,5,6]]`

Output: `[[1,4],[2,5],[3,6]]`

- Constraints:**
- `m == matrix.length`
 - `n == matrix[i].length`
 - `1 <= m, n <= 1000`
 - `1 <= m * n <= 106`

4.3K 133 83 Online

Code Note

```
1 class Solution {
2     public int[][] transpose(int[][] matrix)
3     {
4         int m = matrix.length;
5         int n = matrix[0].length;
6
7         int[][] result = new int[n][m];
8
9         for (int i = 0; i < m; i++) {
10             for (int j = 0; j < n; j++) {
11                 result[j][i] = matrix[i][j];
12             }
13         }
14
15         return result;
16     }
17 }
```

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input

matrix =

[[1,2,3],[4,5,6],[7,8,9]]

Output

1314. Matrix Block Sum

Medium Topics Companies Hint

Given a $m \times n$ matrix `mat` and an integer `k`, return a matrix `answer` where each `answer[i][j]` is the sum of all elements `mat[r][c]` for:

- $i - k \leq r \leq i + k$,
- $j - k \leq c \leq j + k$, and
- (r, c) is a valid position in the matrix.

Example 1:

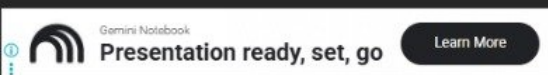
Input: `mat = [[1,2,3],[4,5,6],[7,8,9]]`, `k = 1`
Output: `[[12,21,16],[27,45,33],[24,39,28]]`

Example 2:

Input: `mat = [[1,2,3],[4,5,6],[7,8,9]]`, `k = 2`
Output: `[[45,45,45],[45,45,45],[45,45,45]]`

Constraints:

- $m == \text{mat.length}$
- $n == \text{mat}[i].\text{length}$
- $1 \leq m, n, k \leq 100$
- $1 \leq \text{mat}[i][j] \leq 100$



```
Java Auto
35 // Calculate rectangle sum
36 answer[i][j] = prefix[r2][c2]
37               - prefix[r1 - 1][c2]
38               - prefix[r2][c1 - 1]
39               + prefix[r1 - 1][c1 - 1];
40
41 }
42
43 return answer;
44 }
45
46
```

Testcase Test Result

Input

mat =
[[1,2,3],[4,5,6],[7,8,9]]

k =
1

Output

[[12,21,16],[27,45,33],[24,39,28]]

Expected

[[12,21,16],[27,45,33],[24,39,28]]

Contribute a testcase

Matrix Layer Rotation

Problem

Submissions

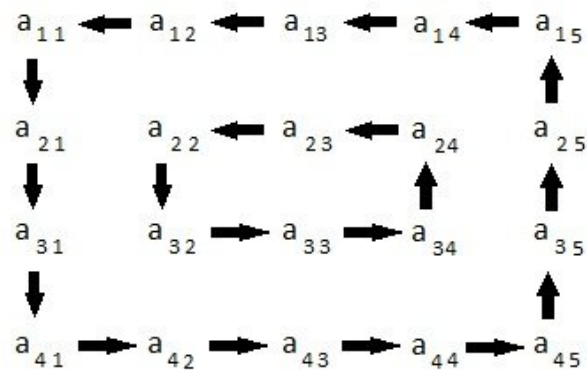
Leaderboard

Discussions

Editorial

You are given a 2D matrix of dimension $m \times n$ and a positive integer r . You have to rotate the matrix r times and print the resultant matrix. Rotation should be in anti-clockwise direction.

Rotation of a 4×5 matrix is represented by the following figure. Note that in one rotation, you have to shift elements by one step only.



Matrix Rotation

It is guaranteed that the minimum of m and n will be even.

As an example rotate the Start matrix by 2:

Start	First	Second
1 2 3 4	2 3 4 5	3 4 5 6
12 1 2 5 ->	1 2 3 6 ->	2 3 4 7

```

5      public static void main(String[] args) {
70          // Top row
71          for (int j = left; j <= right; j++) {
72              matrix[top][j] = rotated.get(index++);
73          }
74
75          // Right column
76          for (int i = top + 1; i <= bottom; i++) {
77              matrix[i][right] = rotated.get(index++);
78          }
79
80          // Bottom row
81          for (int j = right - 1; j >= left; j--) {
82              matrix[bottom][j] = rotated.get(index++);
83          }
84
85          // Left column
86          for (int i = bottom - 1; i > top; i--) {
87              matrix[i][left] = rotated.get(index++);
88          }
89      }
90
91      // Print matrix
92      for (int i = 0; i < m; i++) {
93          for (int j = 0; j < n; j++) {
94              System.out.print(matrix[i][j]);
95
96              if (j < n - 1) {
97                  System.out.print(" ");
98              }
99          }
100         System.out.println();
101     }
102
103     sc.close();
104 }
105

```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

✓ Sample Test case 1

✓ Sample Test case 2

✓ Sample Test case 3

Input (stdin)

```
1 4 4 1
2 1 2 3 4
3 5 6 7 8
4 9 10 11 12
5 13 14 15 16
```

Your Output (stdout)

```
1 2 3 4 8
2 1 7 11 12
3 5 6 10 16
4 9 13 14 15
```